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from sklearn.tree import DecisionTreeClassifier
model = DecisionTreeClassifier(
    max_depth=3,
    min_samples_leaf = 5
)

from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier
from sklearn import tree

# Load Dataset
iris = load_iris()
x = iris.data
y = iris.target

# Create Decision tree model
dt = DecisionTreeClassifier(
    criterion = "gini",      # can also use "entropy"
    max_depth = 3           # pruning (pre-pruning)
)
# train model
dt.fit(x,y)

# prediction
print(dt.predict([[5.1,3.5,1.4,0.2]]))

[0]

from sklearn.datasets import load_iris

from sklearn.ensemble import RandomForestClassifier

import pandas as pd

import numpy as np

# Load dataset
iris = load_iris()

X = iris.data

y = iris.target

# Create Random Forest model
rf = RandomForestClassifier(
    n_estimators=100,      # number of trees
    oob_score=True,        # enable Out-of-Bag error

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    random_state=42
)
# Train model
rf.fit(X, y)
# 1. OOB Score (Out-of-Bag Accuracy)
print("OOB Accuracy:", rf.oob_score_)
print("OOB Error:", 1 - rf.oob_score_)
# 2. Feature Importance
importance = pd.DataFrame({
    "Feature": iris.feature_names,
    "Importance": rf.feature_importances_
})
importance = importance.sort_values(by="Importance", ascending=False)
print("\nFeature Importance:")
print(importance)
# 3. Individual Prediction Example
sample = np.array([[5.1, 3.5, 1.4, 0.2]])
print("\nPrediction:", rf.predict(sample))
OOB Accuracy: 0.9533333333333334
OOB Error: 0.0466666666666666634

Feature Importance:
      Feature  Importance
2  petal length (cm)    0.436130
3  petal width (cm)     0.436065
0  sepal length (cm)    0.106128
1  sepal width (cm)     0.021678

Prediction: [0]

from sklearn.svm import SVC
model = SVC(kernel = "Linear")

```

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# 1. IMPORT LIBRARIES
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix,
    roc_auc_score
)
import numpy as np
# 2. LOAD DATASET
iris = load_iris()
X = iris.data
y = iris.target
# 3. TRAIN-TEST SPLIT
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
# 4. CREATE SVM MODEL
# kernel options: 'linear', 'rbf', 'poly', 'sigmoid'
svm = SVC(
    kernel='rbf',          # non-linear kernel
    probability=True,      # needed for ROC-AUC
    random_state=42
)
# 5. TRAIN MODEL
svm.fit(X_train, y_train)
# 6. PREDICTION
y_pred = svm.predict(X_test)
# probability scores (for ROC-AUC)
y_prob = svm.predict_proba(X_test)
# 7. EVALUATION METRICS
print("MODEL PERFORMANCE")
# Accuracy
acc = accuracy_score(y_test, y_pred)
print("Accuracy:", acc)
# Precision
precision = precision_score(y_test, y_pred, average='macro')
print("Precision:", precision)
# Recall
recall = recall_score(y_test, y_pred, average='macro')
print("Recall:", recall)
# F1 Score
f1 = f1_score(y_test, y_pred, average='macro')
print("F1 Score:", f1)

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# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:\n", cm)
# ROC-AUC (multi-class)
roc = roc_auc_score(y_test, y_prob, multi_class='ovr') #One-vs-Rest
approach
print("\nROC-AUC Score:", roc)
# 8. SINGLE PREDICTION
sample = np.array([[5.1, 3.5, 1.4, 0.2]])
prediction = svm.predict(sample)
print("\nSingle Sample Prediction:", prediction)

```

#### MODEL PERFORMANCE

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Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0

```

#### Confusion Matrix:

```

[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

```

ROC-AUC Score: 1.0

Single Sample Prediction: [0]

#### # 1. IMPORT LIBRARIES

```

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from sklearn.model_selection import train_test_split

from sklearn.svm import SVC

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    roc_auc_score
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```

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X = iris.data
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X_train, X_test, y_train, y_test = train_test_split(
    X, y,
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```

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recall = recall_score(y_test, y_pred, average='macro')
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f1 = f1_score(y_test, y_pred, average='macro')
print("F1 Score:", f1)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:\n", cm)

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roc = roc_auc_score(y_test, y_prob, multi_class='ovr') #One-vs-Rest
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print("\nROC-AUC Score:", roc)

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sample = np.array([[5.1, 3.5, 1.4, 0.2]])
prediction = svm.predict(sample)
print("\nSingle Sample Prediction:", prediction)

MODEL PERFORMANCE
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0

Confusion Matrix:

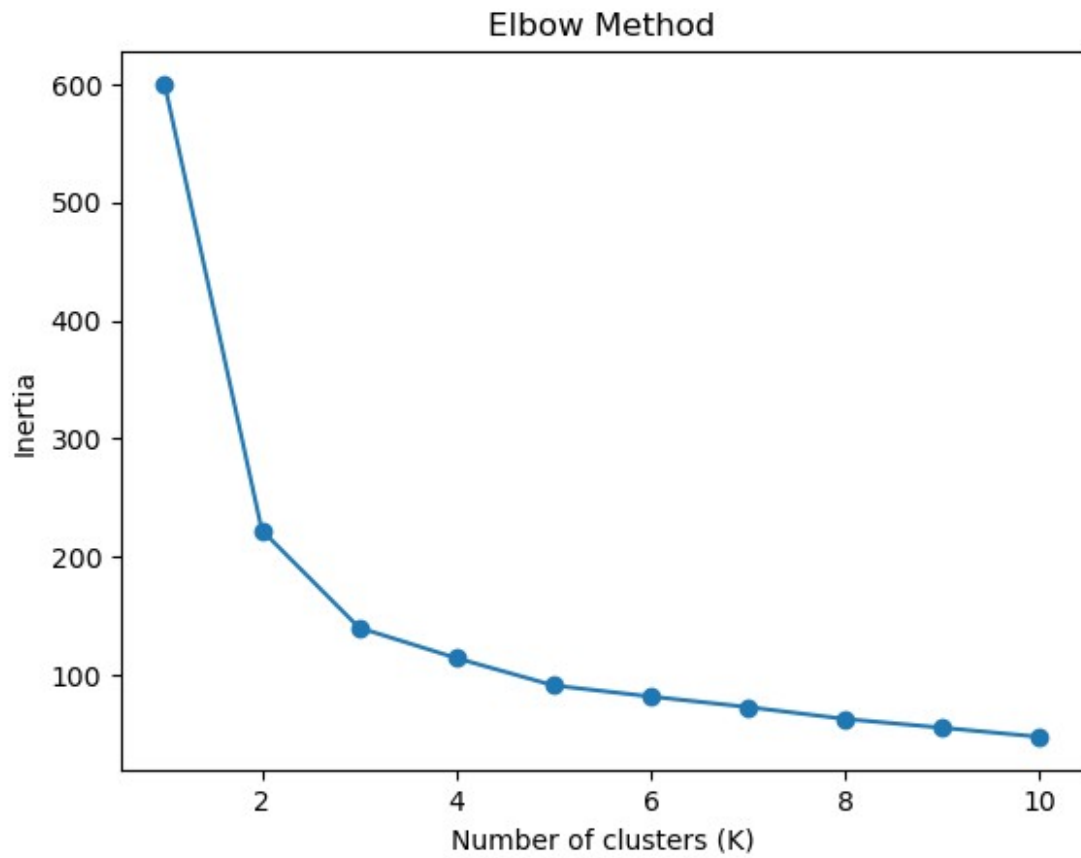
```

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

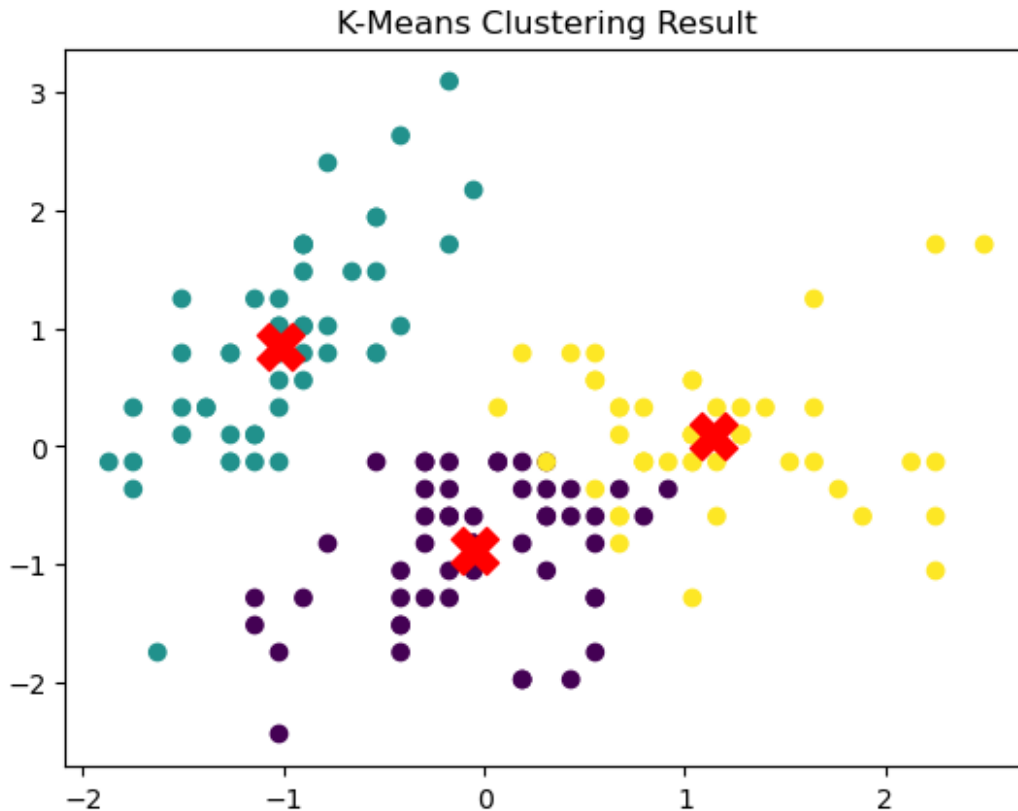
ROC-AUC Score: 1.0

Single Sample Prediction: [0]

```
import warnings
warnings.filterwarnings("ignore")
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.preprocessing import StandardScaler
# 1. Load dataset
iris = load_iris()
X = iris.data
# 2. Feature scaling (important for K-Means)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
# 3. Elbow Method (Inertia)
inertia = []
K_range = range(1, 11)
for k in K_range:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    inertia.append(kmeans.inertia_)
plt.plot(K_range, inertia, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of clusters (K)")
plt.ylabel("Inertia")
plt.show()
# 4. Fit final KMeans (choose K = 3 for iris)
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
labels = kmeans.fit_predict(X_scaled)
# 5. Silhouette Score
score = silhouette_score(X_scaled, labels)
print("Silhouette Score:", score)
# 6. Visualize clusters (first 2 features)
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=labels, cmap='viridis')
plt.scatter(kmeans.cluster_centers_[:, 0],
            kmeans.cluster_centers_[:, 1],
            s=300, c='red', marker='X')
plt.title("K-Means Clustering Result")
plt.show()
```



Silhouette Score: 0.45994823920518635



```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from scipy.cluster.hierarchy import dendrogram, linkage
from sklearn.cluster import AgglomerativeClustering

# Load dataset
iris = load_iris()
X = iris.data

# Scale features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Create linkage matrix
linked = linkage(X_scaled, method='ward')

# Plot dendrogram
plt.figure(figsize=(10,5))
dendrogram(linked)
plt.title("Dendrogram")
plt.xlabel("Samples")
plt.ylabel("Distance")
plt.show()

# Hierarchical Clustering
hc = AgglomerativeClustering(
    n_clusters=3,
    linkage='ward'
```

